

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

University Examination

Date: 05/06/2018

Day: Tuesday

Time: 10 am to 10:30 am

Total Marks: 20

Student's ID No.									
Student's Name: _____									
Student's Sign : _____									
Supervisor's Name : _____									
Supervisor's Sign: _____									

Answer Sheet for Multiple Choice Questions (MCQs), True or false and Match the following
Examination : BPT/MPT- June 2018
Semester: 2 nd
Section : I
Course Code : PT791
Course Name: Sports Biomechanics and Performance Analysis

Important Instructions:

- There are two sections in the paper. **SECTION –I** contains 20 Questions and **SECTION – II** contains 3 Questions.
- **SECTION –I** should be answered in the question paper itself & **SECTION – II** should be answered in separate answer book.
- Marks on the right side suggest the total marks of that question.
- Maximum time allotted for **SECTION – I** is 30 minutes Question paper for **SECTION – II** will be given once the **SECTION – I** is completed within the stipulated time and returned. However, Candidates who complete **SECTION – I** earlier than 30 minutes can collect main question paper for **SECTION –II** immediately after handing over the **SECTION-I** question paper with answers.
- **SECTION – I** should be attached with the main answer book at the end of the examination and should be submitted to the supervisor.

SECTION-I

Multiple Choice Questions: (MCQs) (Encircle the correct answer)

(10 × 1 = 10)

1. Following details are used to describe motion, except _____
 - a. Time
 - b. Position
 - c. Displacement
 - d. Vectors
2. Patellar loading while descending stairs is _____
 - a. 0.3 times the body weight
 - b. 2.5 times the body weight
 - c. 3.5 times the body weight
 - d. 7 times the body weight
3. Which of the following mode of resistance does not affect by gravitational force?
 - a. Sand bag
 - b. Medicine ball
 - c. Theraband
 - d. Dumbbells

4. Objective of 1-RM bench-press test is to test strength of following muscles except ____
 - a. Triceps
 - b. Pectoralis major
 - c. Biceps
 - d. Anterior deltoid
5. Which of the following is not a normal patterns of foot strikes in running?
 - a. Lateral heel strike
 - b. Medial-foot strike
 - c. Mid-foot strike
 - d. Forefoot strike
6. Which of the following muscle exercise regimes is not used in plyometric training?
 - a. Concentric
 - b. Eccentric
 - c. Isokinetic
 - d. Isometric
7. Following are the reasons for measuring body composition, except _____
 - a. To assess the metabolic changes before weight-management program.
 - b. To help athletes determine the best body composition for performance.
 - c. To monitor fat & fat-free weight in disease pt.
 - d. To track long-term changes that occur in body fat and fat-free mass with aging.
8. Striking a golf requires following principles, except _____
 - a. Newtons laws (force)
 - b. Friction
 - c. Momentum
 - d. Stability
9. Which of the following test is not for testing speed of an athletes?
 - a. Running speed test
 - b. Winagate test
 - c. Sprint test
 - d. 30 yard dash test
10. A standard zig zag test requires a rectangular area of _____
 - a. 10 / 16 feet
 - b. 12 / 18 feet
 - c. 14 / 20 feet
 - d. 16 / 22 feet

Identify the sentence as True or False:

(5 × 1 = 5)

(Please tick mark the answer) True False

11. Body composition only refers to the proportion of body weight.
12. Pressure = F / N
13. Rectilinear motion is movement that occurs in a straight line.
14. Kinematic explains the cause of motion
15. Whiplash injuries is an example of Newton's 2nd law

☐	☐
☐	☐
☐	☐
☐	☐
☐	☐

(P.T.O.)

Match the following**(5 × 1 = 5)**

16.	Push or pull that tends to produce acceleration	A. Speed
17.	Ability to change direction rapidly and with accuracy	B. Stability
18.	Ability to perform movement in a short time as possible	C. Force
19.	Resistance to overthrowing or sudden changes.	D. Equilibrium
20.	Balance point of an object at which torque on all sides is equal	E. Agility
		F. Center of gravity
		G. Power

(16 -), (17-), (18-), (19-), (20-)

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

2nd Semester of M.P.T. Examination

June 2018

PT791 SPORTS BIOMECHANICS AND PERFORMANCE ANALYSIS

Date: 05/06/2018

Time: 10:30 – 1 pm

Maximum Marks: 50

Instructions:

1. There are two Sections in the paper. **SECTION –I** contains 20 Questions and **SECTION – II** contains 3 Questions.
2. **SECTION –I** should be answered in the question paper itself & **SECTION – II** should be answered in separate answer book.
3. Marks on the right side suggest the total marks of that question.
4. Maximum time allotted for **SECTION – II** is 2 Hr 30 min.
5. Draw the figure where necessary.

SECTION-II

Q.1. Short answers (5 out of 7) (Answer not exceeding 50 words) (5 × 2 = 10)

- a. Fluid forces
- b. Types of motion
- c. Windlass mechanism
- d. Accelerometer
- e. Side-step test and its scoring
- f. Factors affecting trajectory
- g. Motion capture technologies

Q.2. Short Notes (4 out of 5) (Answer not exceeding 150 words) (4 × 5 = 20)

- a. Curl-up muscular endurance test
- b. Explain about agility and write in brief about any one of the agility testing.
- c. Bioelectrical Impedance Analysis (BIA)
- d. Enumerate balance test and discuss a dynamic balance test
- e. Force Platforms

Q.3. Essay (2 out of 3) (2 × 10 = 20)

- a. Describe about phases of throwing and brief the injuries associated with throwing.
- b. Explain in detail about the muscles involved while playing badminton and brief the possible injuries.
- c. Describe about biomechanics of running and brief the injuries associated with running.